CS2601 Linear and Convex Optimization

Homework 6

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1. (a.)The barrier function is $B(\boldsymbol{x}) = -\sum\limits_{i=1}^{n} logx_i$

   The approximating equality constrained problem is :

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{c}^T\boldsymbol{x} - \frac{1}{t}\sum_{i=1}^{n} log(x_i) \\ s.t. \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \end{aligned}$$

   (b.)Let $g(\boldsymbol{x}) = \boldsymbol{c}^T\boldsymbol{x} - \frac{1}{t}\sum\limits_{i=1}^{n} log(x_i)$

   $\nabla g(\boldsymbol{x}) = \boldsymbol{c} - \frac{1}{t}[\frac{1}{x_1}, \frac{1}{x_2}, ..., \frac{1}{x_n}]^T$

   $\nabla^2 g(\boldsymbol{x}) = \frac{1}{t}\begin{bmatrix} \frac{1}{x_1^2} & 0 & ... & 0 \\ 0 & \frac{1}{x_2^2} & ... & 0 \\ 0 & 0 & ... & \frac{1}{x_n^2} \end{bmatrix}$

   (c.)

```
import newton as nt
import numpy as np
from copy import deepcopy
def centering_step(c, A, b, x0, t):
    def f(x):
        return c.T @ x - 1.0/t * sum(np.log(np.maximum(x, 1e-10)))

    def fp(x):
        return c - 1.0/(t * x)

    def fpp(x):
        return np.diag((1.0/(x**2 * t)).T[0])
    return nt.newton_eq(f, fp, fpp, x0, A, b)[-1]

def barrier(c, A, b, x0, tol=1e-8, t0=1, rho=10):

    t = t0
    x = np.array(x0)
    x_traces = [np.array(x0)]

    for it in range(1000):
        x1 = deepcopy(x)
        x = centering_step(c, A, b, x1, t)
        x_traces.append(deepcopy(x))
        if (1 / t) <= tol:
                break
        t = rho * t
    return x_traces
```

(d.)Convert the LP problem to standard form:

$$\begin{aligned} &\min_{\boldsymbol{x}\in\mathbb{R}^4} \quad -x_1 - 3x_2 \\ &s.t. \quad x_1 + x_2 + x_3 = 6 \\ &-x_1 + 2x_2 + x_4 = 8 \\ &x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Let $\boldsymbol{c} = [-1, -3, 0, 0]^T$, $\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$

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iteration 0: [[3.]
 [1.]
 [2.]
 [9.]]
iteration 1: [[1.63307563]
 [3.90402064]
 [0.46290373]
 [1.82503435]]
iteration 2: [[1.34600004]
 [4.59597677]
 [0.05802319]
 [0.1540465 ]]
iteration 3: [[1.3343604 ]
 [4.65966009]
 [0.00597951]
 [0.01504022]]
iteration 4: [[1.33343360e+00]
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 [5.99794358e-04]
 [1.50040181e-03]]
```

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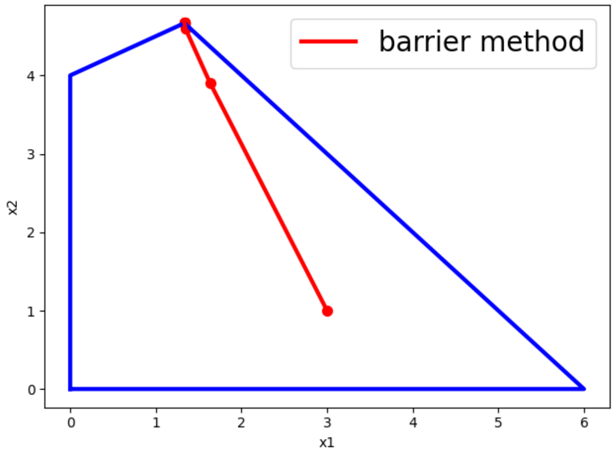
iteration 5: [[1.33334334e+00]
 [4.66659667e+00]
 [5.99948876e-05]
 [1.49996563e-04]]
iteration 6: [[1.33333433e+00]
 [4.66665967e+00]
 [5.99937452e-06]
 [1.49985317e-05]]
iteration 7: [[1.33333343e+00]
 [4.66666597e+00]
 [5.99939244e-07]
 [1.49984906e-06]]
iteration 8: [[1.33333334e+00]
 [4.66666660e+00]
 [5.93972887e-08]
 [1.48493250e-07]]
iteration 9: [[1.33333333e+00]
 [4.66666666e+00]
 [3.99866757e-09]
 [9.99667352e-09]]
optimal value: [[-15.33333332]]
```



2. (a).

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu}) = -x_1 - 3x_2 + \mu_1(x_1 + x_2 - 6) + \mu_2(-x_1 + 2x_2 - 8) + \mu_3(-x_1) + \mu_4(-x_2)$$
$$= (-1 + \mu_1 - \mu_2 - \mu_3)x_1 + (-3 + \mu_1 + 2\mu_2 - \mu_4)x_2 - 6\mu_1 - 8\mu_2$$

Let $\boldsymbol{A} = \begin{bmatrix} 1 & -1 & -1 & 0 \\ 1 & 2 & 0 & -1 \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \end{bmatrix}$

The dual function is $\phi(\boldsymbol{\mu}) = \inf_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu}) = \begin{cases} -6\mu_1 - 8\mu_2, & \boldsymbol{A\mu} = \boldsymbol{b} \\ -\infty, & \boldsymbol{A\mu} \neq \boldsymbol{b} \end{cases}$

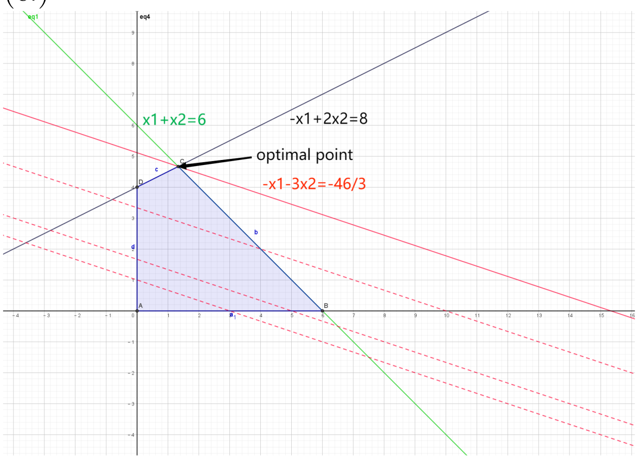
The dual LP in standard form is

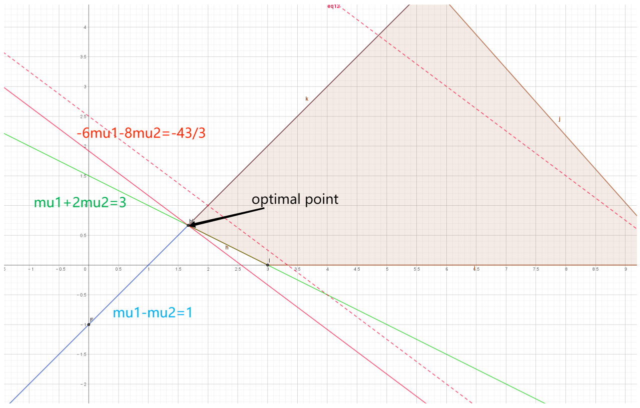
$$
\begin{aligned}
&\max_{\boldsymbol{\mu}\in\mathbb{R}^4} \quad \psi(\boldsymbol{\mu}) = -6\mu_1 - 8\mu_2 \\
&s.t. \quad \mu_1 - \mu_2 - \mu_3 = 1 \\
&\mu_1 + 2\mu_2 - \mu_4 = 3 \\
&\mu_1, \mu_2, \mu_3, \mu_4 \geq 0
\end{aligned}
$$

(b.)The symmetric dual LP is:

$$
\begin{aligned}
&\max_{\boldsymbol{\mu}\in\mathbb{R}^4} \quad \psi(\boldsymbol{\mu}) = -6\mu_1 - 8\mu_2 \\
&s.t. \quad -\mu_1 + \mu_2 \leq 1 \\
&-\mu_1 - 2\mu_2 \leq 3 \\
&\mu_1, \mu_2 \geq 0
\end{aligned}
$$

(c.)





Therefore $f^* = -43/3, \psi^* = -43/3, f^* = \psi^*$

(d.)

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iteration 0: [[4.]
 [1.]
 [2.]
 [3.]]
iteration 1: [[2.1602764 ]
 [0.54793489]
 [0.61234151]
 [0.25614619]]
iteration 2: [[1.72344885]
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 [0.0742942 ]
 [0.02175816]]
iteration 3: [[1.67237818]
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 [0.00749423]
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iteration 4: [[1.66723807e+00]
 [6.66488125e-01]
 [7.49943594e-04]
 [2.14317863e-04]]
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iteration 5: [[1.66672380e+00]
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iteration 6: [[1.66667238e+00]
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 [7.49923738e-06]
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iteration 7: [[1.66666723e+00]
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 [7.42466015e-07]
 [2.12133296e-07]]
iteration 8: [[1.66666672e+00]
 [6.66666651e-01]
 [6.66555489e-08]
 [1.90444589e-08]]
iteration 9: [[1.66666667e+00]
 [6.66666666e-01]
 [9.38122456e-10]
 [2.68017088e-10]]
dual optimal value: [[-15.33333334]]
```

3. (a.)$f'(x) = \frac{e^x}{1+e^x} > 0$
$\therefore f(x)$ is monotonically increasing.
$\therefore$The optimal point is 0 and the optimal value is $f^*(x) = f(0) = log2$
(b.)$\mathcal{L}(x, \mu) = log(1 + e^x) - \mu x$
The dual function is

$$\phi(\mu) = \inf_x [log(1 + e^x) - \mu x]$$

If $\mu \leq 0$, $\frac{\partial \mathcal{L}}{\partial x} = \frac{e^x}{1+e^x} - \mu > 0$
Therefore $\phi(\mu) = lim_{x \to -\infty} \mathcal{L}(x, \mu) = \begin{cases} 0, & \mu = 0 \\ -\infty, & \mu < 0 \end{cases}$
If $\mu \geq 1$, $\frac{\partial \mathcal{L}}{\partial x} = \frac{e^x}{1+e^x} - \mu < 0$
Therefore $\phi(\mu) = lim_{x \to +\infty} \mathcal{L}(x, \mu) = \begin{cases} 0, & \mu = 1 \\ -\infty & \mu > 1 \end{cases}$
If $0 < \mu < 1$, let $\frac{\partial \mathcal{L}}{\partial x} = \frac{e^x}{1+e^x} = 0$, then $x = log\frac{\mu}{1-\mu}$
$\phi(\mu) = -\mu log\mu - (1 - \mu) log(1 - \mu)$
Therefore, the dual function is

$$\phi(\mu) = \begin{cases} -\mu log\mu - (1 - \mu) log(1 - \mu), & 0 < \mu < 1 \\ 0, & \mu = 1 or \mu = 0 \\ -\infty & \mu > 1 or \mu < 0 \end{cases}$$

Therefore the dual problem is

(c.)When $0 < \mu < 1$, let $\phi'(\mu) = log\frac{1-\mu}{\mu} = 0$
$\therefore \mu = \frac{1}{2}, \phi(\frac{1}{2}) = log2 > 0$
$\therefore$The dual optimal point is $\mu^* = \frac{1}{2}$ and the dual optimal value is $\phi^* = log2 = f^*$
The strong duality hold.

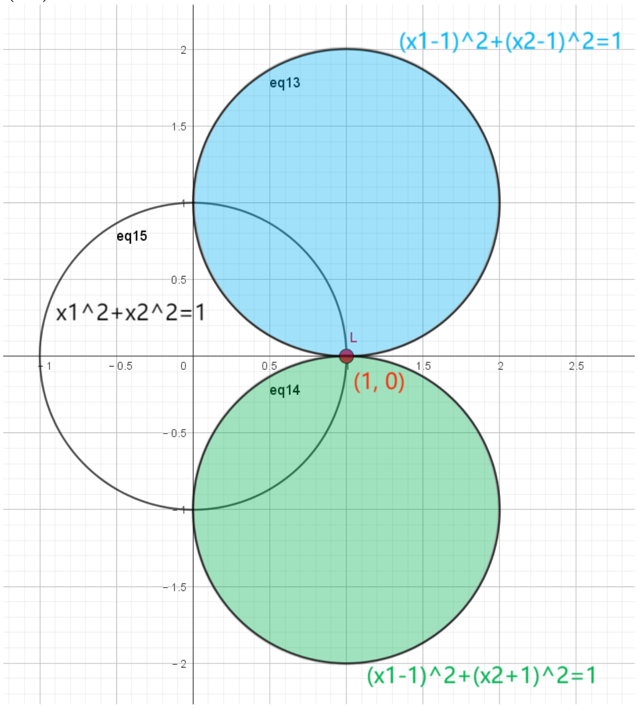
4. (a.)$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu}) = x_1^2 + x_2^2 + \mu_1((x_1 - 1)^2 + (x_2 - 1)^2 - 1) + \mu_2((x_1 - 1)^2 + (x_2 + 1)^2 - 1) = (1 + \mu_1 + \mu_2)x_1^2 + (1 + \mu_1 + \mu_2)x_2^2 - 2(\mu_1 + \mu_2)x_1 + 2(\mu_2 - \mu_1)x_2 + \mu_1 + \mu_2$
If $\mu_1 + \mu_2 \leq -1$, $\mathcal{L} \to -\infty$ when $x_1 \to +\infty, x_2 \to +\infty$
Therefore $\phi(\boldsymbol{\mu}) = \inf_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu}) = -\infty$
If $\mu_1 + \mu_2 > -1$, $\nabla \mathcal{L} = [2x_1 + 2\mu_1(x_1 - 1) + 2\mu_2(x_1 - 1), 2x_2 + 2\mu_1(x_2 - 1) + 2\mu_2(x_2 + 1)]^T$
Let $\nabla \mathcal{L} = 0$,
$\begin{cases} x_1 = \frac{\mu_1 + \mu_2}{1 + \mu_1 + \mu_2} \\ x_2 = \frac{\mu_1 - \mu_2}{1 + \mu_1 + \mu_2} \end{cases}$
Therefore $\phi(\boldsymbol{\mu}) = \inf_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{\mu}) = \frac{\mu_1 + \mu_2 - (\mu_1 - \mu_2)^2}{1 + \mu_1 + \mu_2}$
The dual function is:

$$\phi(\boldsymbol{\mu}) = \begin{cases} \frac{\mu_1 + \mu_2 - (\mu_1 - \mu_2)^2}{1 + \mu_1 + \mu_2}, & \mu_1 + \mu_2 > -1 \\ -\infty, & \mu_1 + \mu_2 \leq -1 \end{cases}$$

The dual problem is

$$\begin{aligned} \underset{\boldsymbol{\mu}\in\mathbb{R}^2}{max} \quad & \phi(\boldsymbol{\mu}) = \begin{cases} \frac{\mu_1+\mu_2-(\mu_1-\mu_2)^2}{1+\mu_1+\mu_2}, & \mu_1+\mu_2 > -1 \\ -\infty, & \mu_1+\mu_2 \le -1 \end{cases} \\ s.t. \quad & \boldsymbol{\mu} \ge 0 \end{aligned}$$

(b.)



From the graph we can see $\boldsymbol{x}^* = (1, 0)$ is the only feasible point.
Therefore $f^* = 1$
$\because \phi(\boldsymbol{\mu}) = \frac{\mu_1+\mu_2-(\mu_1-\mu_2)^2}{1+\mu_1+\mu_2} \le \frac{\mu_1+\mu_2}{1+\mu_1+\mu_2} < 1 = f^*$
Let $\mu_1 = \mu_2, \phi(\boldsymbol{\mu}) = \frac{2\mu_1}{1+2\mu_1}$. When $\mu_1 \to +\infty$, $\phi(\boldsymbol{\mu}) \to 1$
$\therefore \phi^* = \underset{\boldsymbol{\mu}\ge\boldsymbol{0}}{sup} \quad \phi(\boldsymbol{\mu}) = 1 = f^*$
Therefore strong duality holds
(c.)$\because$For the only feasible point $\boldsymbol{x}^* = (1, 0)$, $(x_1-1)^2+(x_2-1)^2-1 = 0$, $(x_1-1)^2+(x_2+1)^2-1 = 0$
$\therefore$ Slater's condition doesn't hold
Therefore Slater's condition is not necessary for strong duality to hold
(d.)If $\phi(\boldsymbol{\mu}) = 1$, $(\mu_1 - \mu_2) = -1$. Impossible
$\therefore$ The dual optimal value ϕ^* is not attained by any dual feasible point.This is expected.